Proprietary + Confidential



Data engineering tasks and components

Proprietary + Confidential

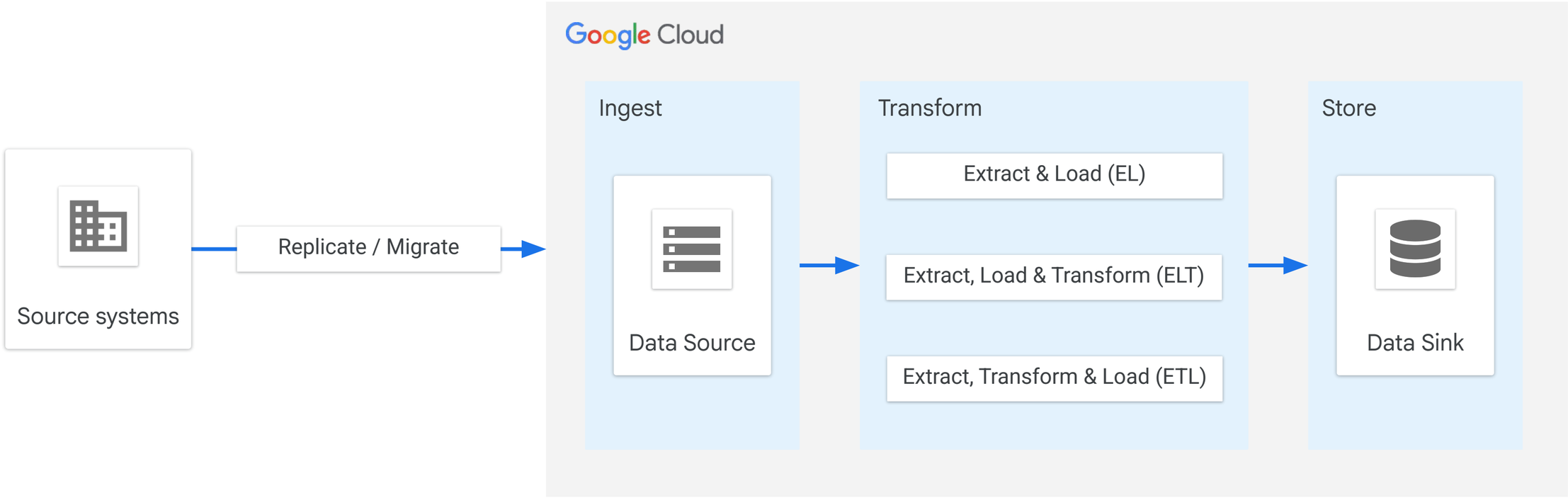
Data engineering tasks and components

01	The role of a data engineer
02	Data sources versus data syncs
03	Data formats
04	Storage solution options on Google Cloud
05	Metadata management options on Google Cloud
06	Share datasets using Analytics Hub



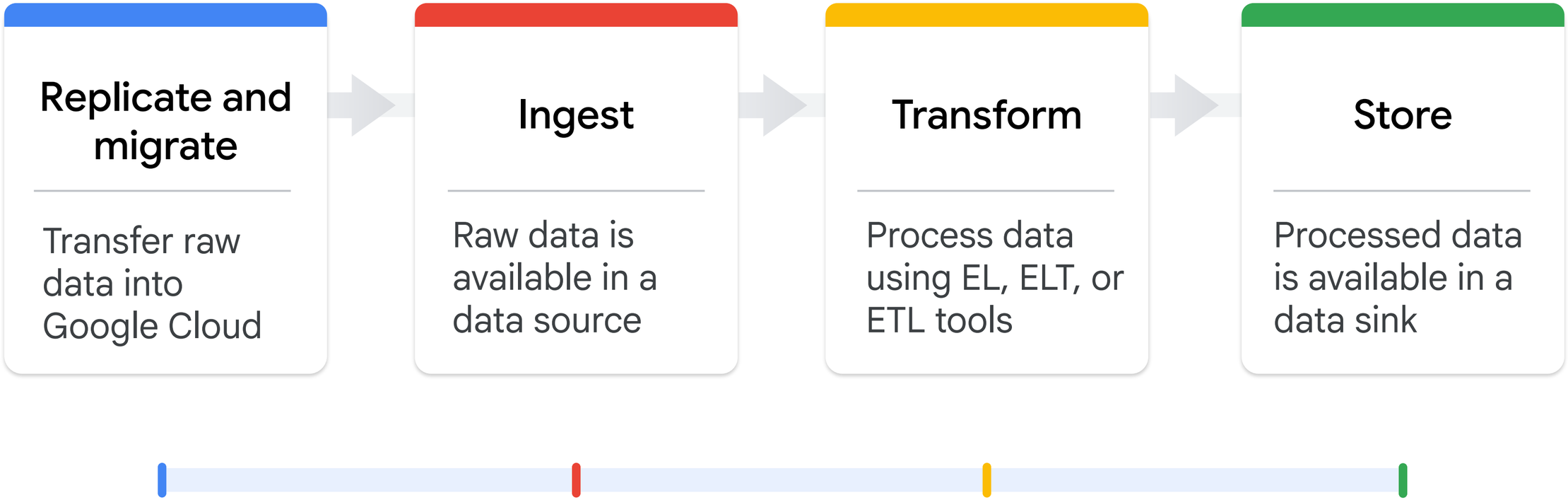
Proprietary + Confidential

Data engineering tasks evolve around ingesting, transforming, and storing data



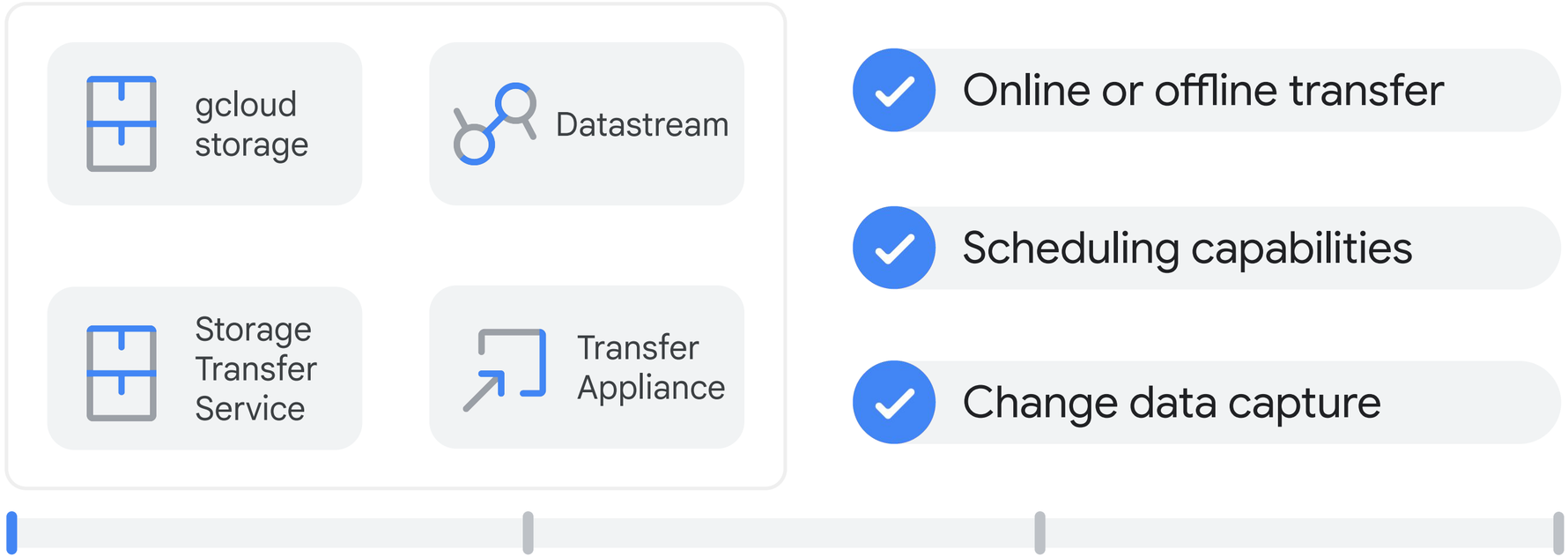
Proprietary + Confidential

Data engineering tasks evolve around ingesting, transforming, and storing data



Proprietary + Confidential

Replication and migration services onboard your data into Google Cloud



Proprietary + Confidential

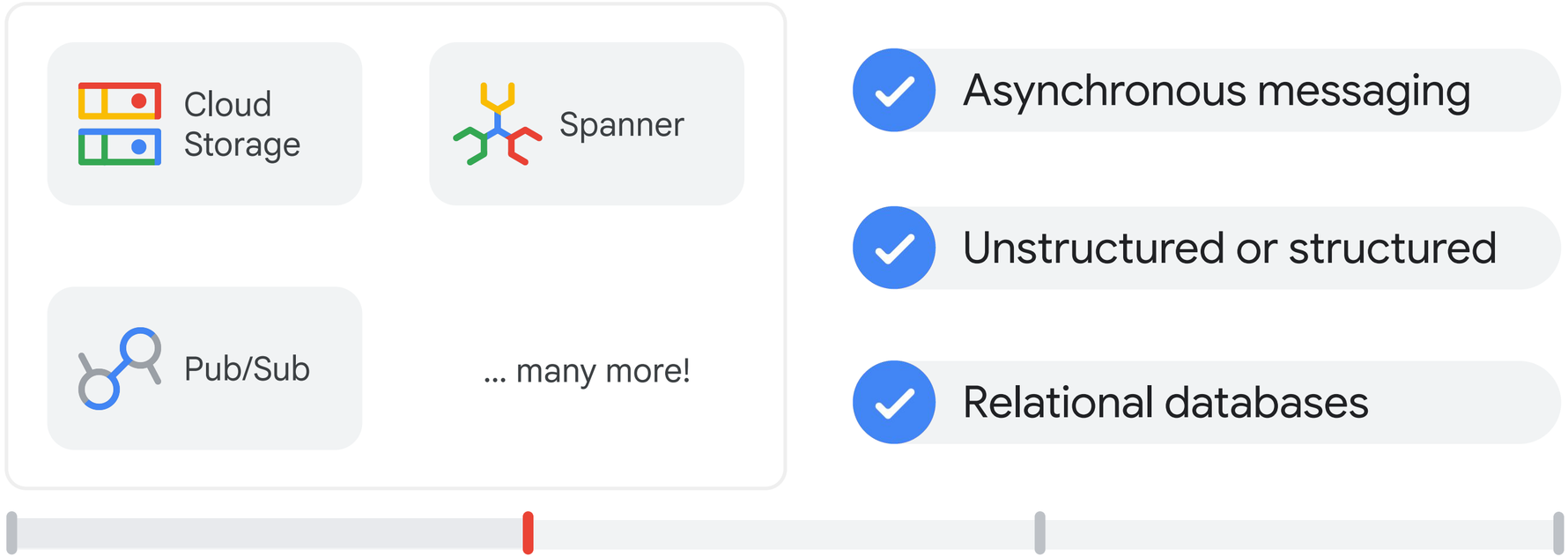
Data engineering tasks and components

01	The role of a data engineer
02	Data sources versus data syncs
03	Data formats
04	Storage solution options on Google Cloud
05	Metadata management options on Google Cloud
06	Share datasets using Analytics Hub



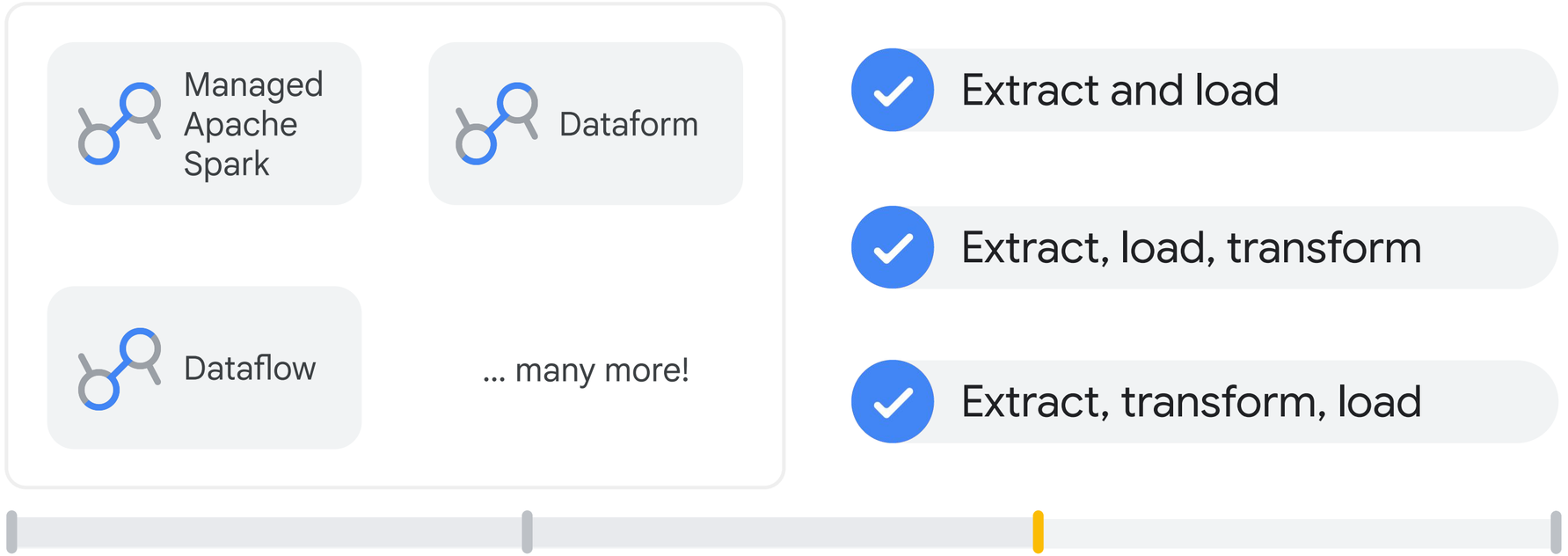
Proprietary + Confidential

Data sources are the origin point of your raw data on Google Cloud



Proprietary + Confidential

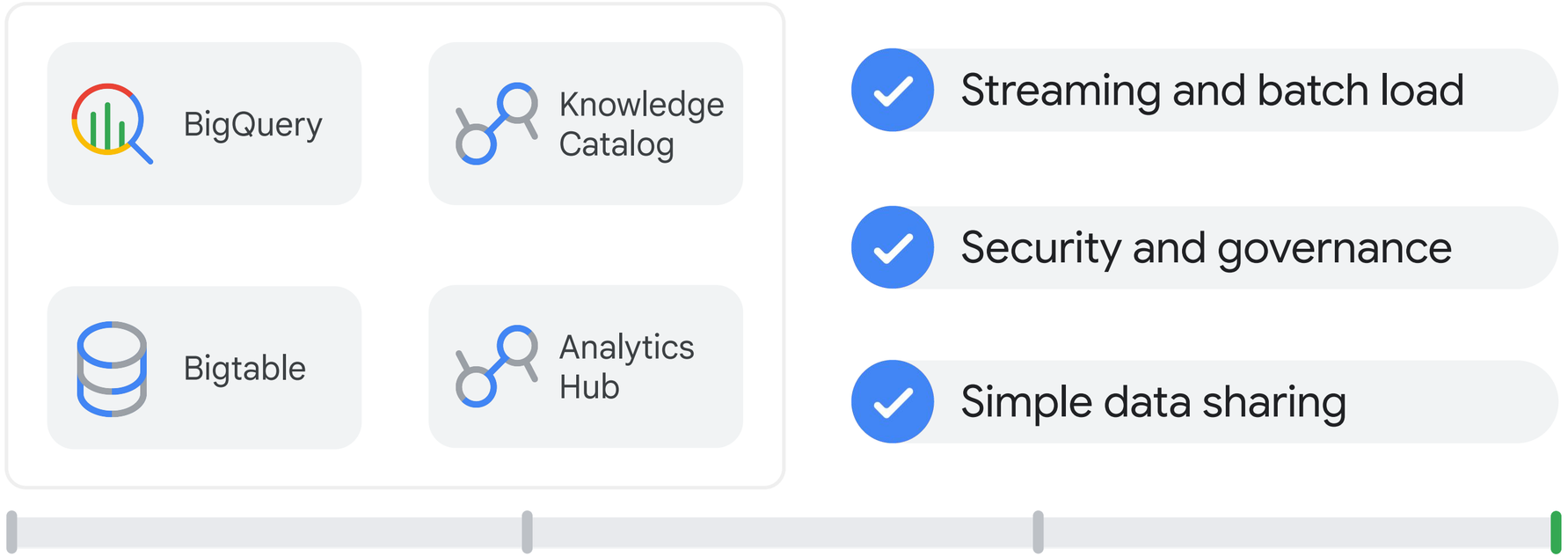
Transformation services add new value to your data



Replicate and migrate | Ingest | **Transform** | Store

Proprietary + Confidential

Data sinks store your processed data on Google Cloud



Proprietary + Confidential

Data engineering tasks and components

01	The role of a data engineer
02	Data sources versus data syncs
03	Data formats
04	Storage solution options on Google Cloud
05	Metadata management options on Google Cloud
06	Share datasets using Analytics Hub



Proprietary + Confidential

Data can have different formats

Storage/process data

- Data type
- Business need

Unstructured



- Documents
- Images
- Audio files

Structured



- Tables
  - Rows
  - Columns

Proprietary + Confidential

Data engineering tasks and components

01	The role of a data engineer
02	Data sources versus data syncs
03	Data formats
04	Storage solution options on Google Cloud
05	Metadata management options on Google Cloud
06	Share datasets using Analytics Hub



Proprietary + Confidential

Cloud Storage holds your unstructured data

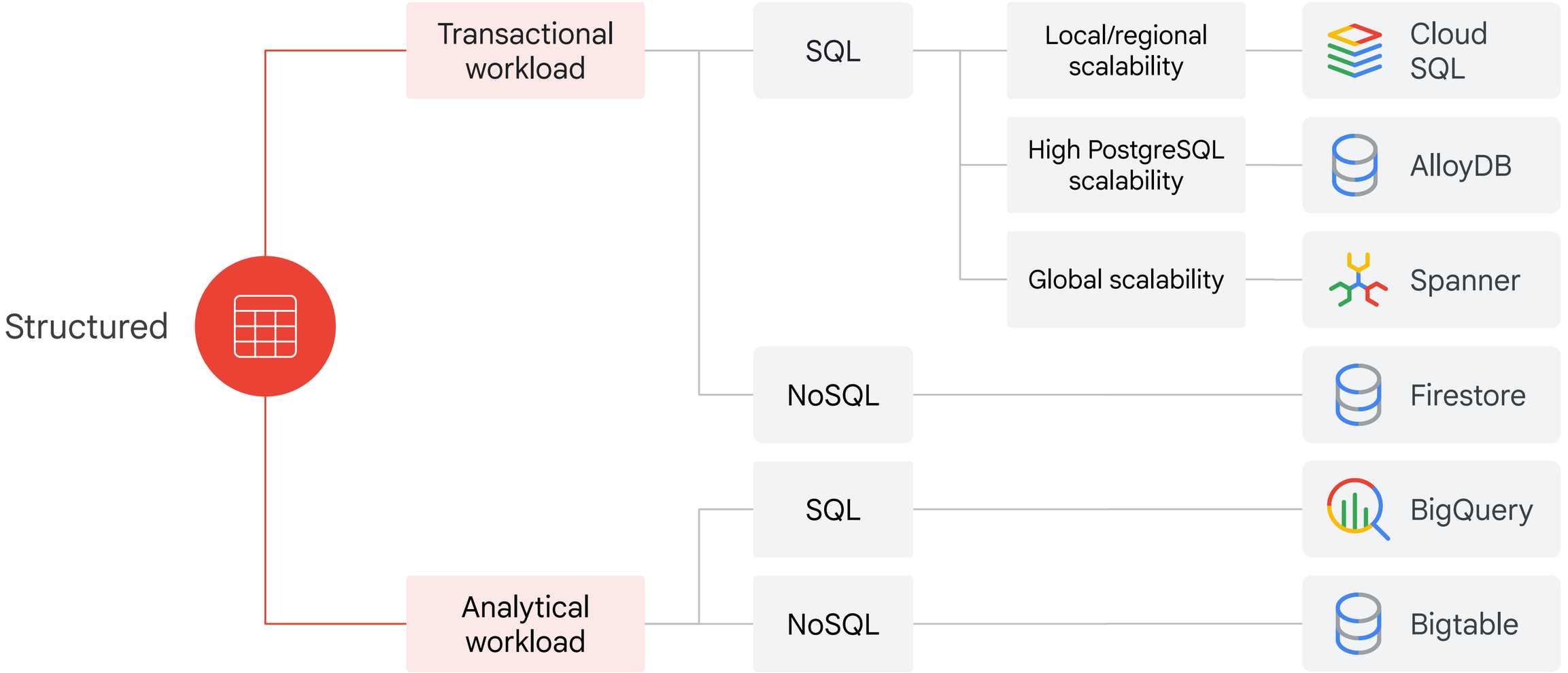
Standard storage	Nearline storage	Coldline storage	Archive storage
			
Hot data	Once per month	Once every 90 days	Once a year

- Application data
- Database backups
- Log files
- Compliance data

- Reliability and scalability
- Accessed by HTTP request
- Retrieved by object name
- Max object size of 5 TB

Proprietary + Confidential

Options for storing structured data



Proprietary + Confidential

Data lake versus data warehouse

	Data lake	Data warehouse
Data format	Native (unstructured, semi-structured, structured)	Schema (structured or semi-structured)
Data type	Raw	Pre-processed and aggregated from multiple data sources
Purpose	Data science, applications, business decisions	Long-term business analysis
Dependencies	Tools/processes to enable data discovery, governance, security, metadata management	Standalone
Service	 Cloud Storage	 BigQuery

Proprietary + Confidential

BigQuery is a serverless, fully managed data warehouse

Security on dataset, table, column and row level

Built-in **machine learning** and **geographic information system**

Scalable storage and analytics services



BigQuery

Rich ecosystem for data transformations

Integration with other storage services

Real-time analytics on streaming data

Proprietary + Confidential

Connecting to BigQuery is easy

RUN

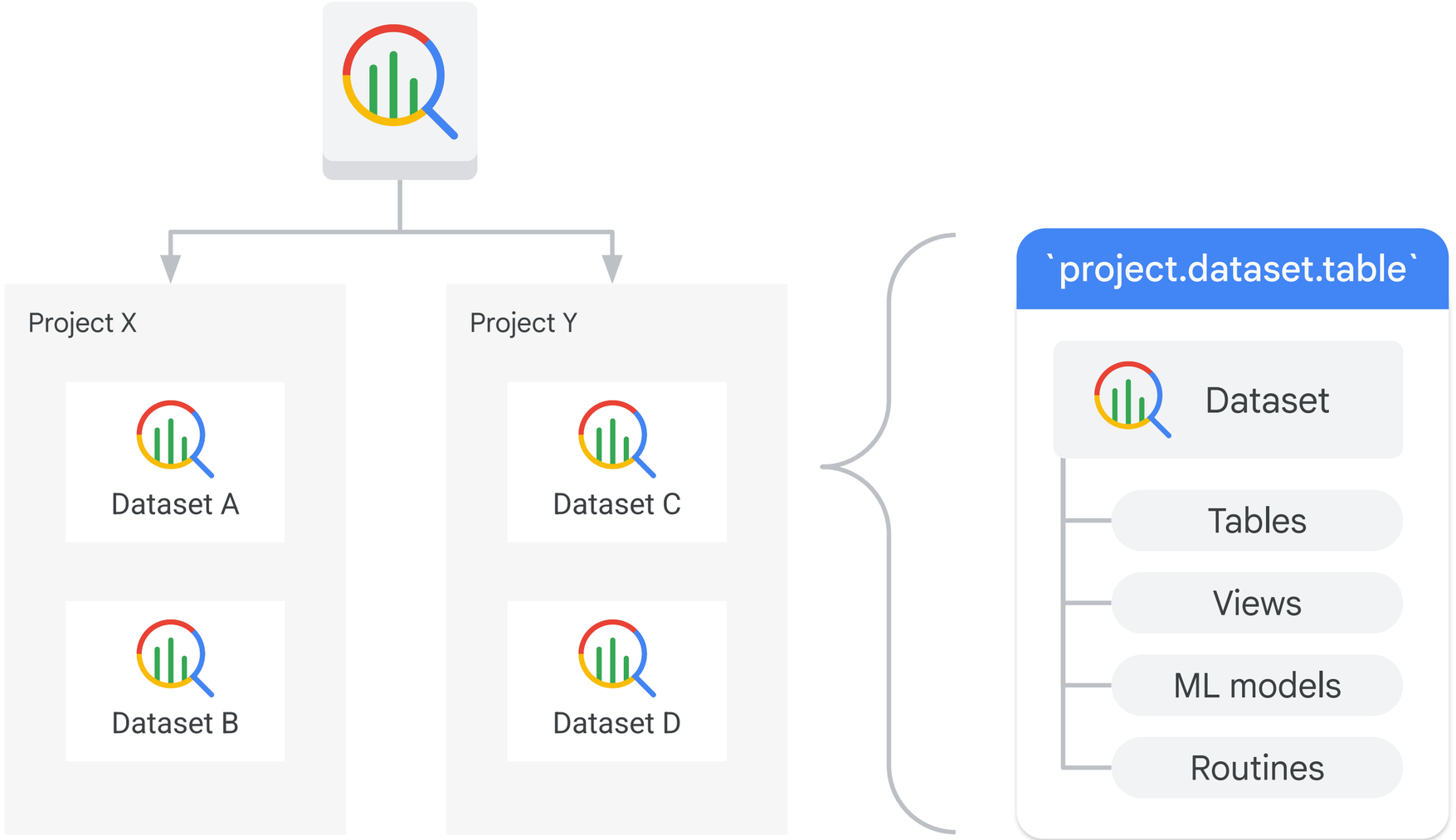
```
# Get count of comments by user for articles
# with 'Google' in the title
SELECT
  contributor_username,
  COUNT(comment) AS comments
FROM
  `bigquery-public-data.samples.wikipedia`
WHERE
  title LIKE "%Google%"
  AND contributor_username IS NOT NULL
GROUP BY 1
ORDER BY 2 DESC
LIMIT 100;
```

```
> bq query --use_legacy_sql=false \
  'SELECT contributor_username, \
  COUNT(comment) AS comments \
  FROM ...;'
```

- Web UI with SQL Editor
- bq command line tool
- REST API (7 languages supported)

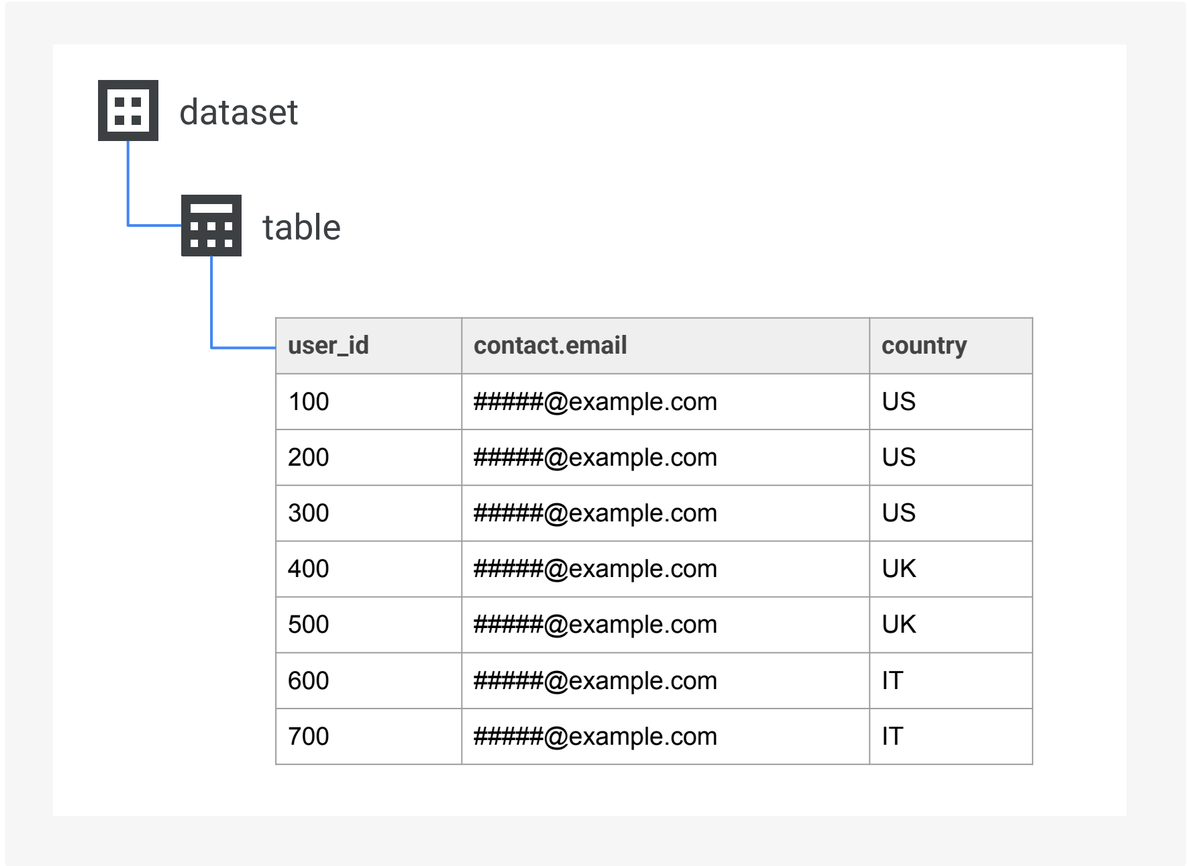
Proprietary + Confidential

How BigQuery's resources are organized



Proprietary + Confidential

You can secure your BigQuery resources on multiple levels



Proprietary + Confidential

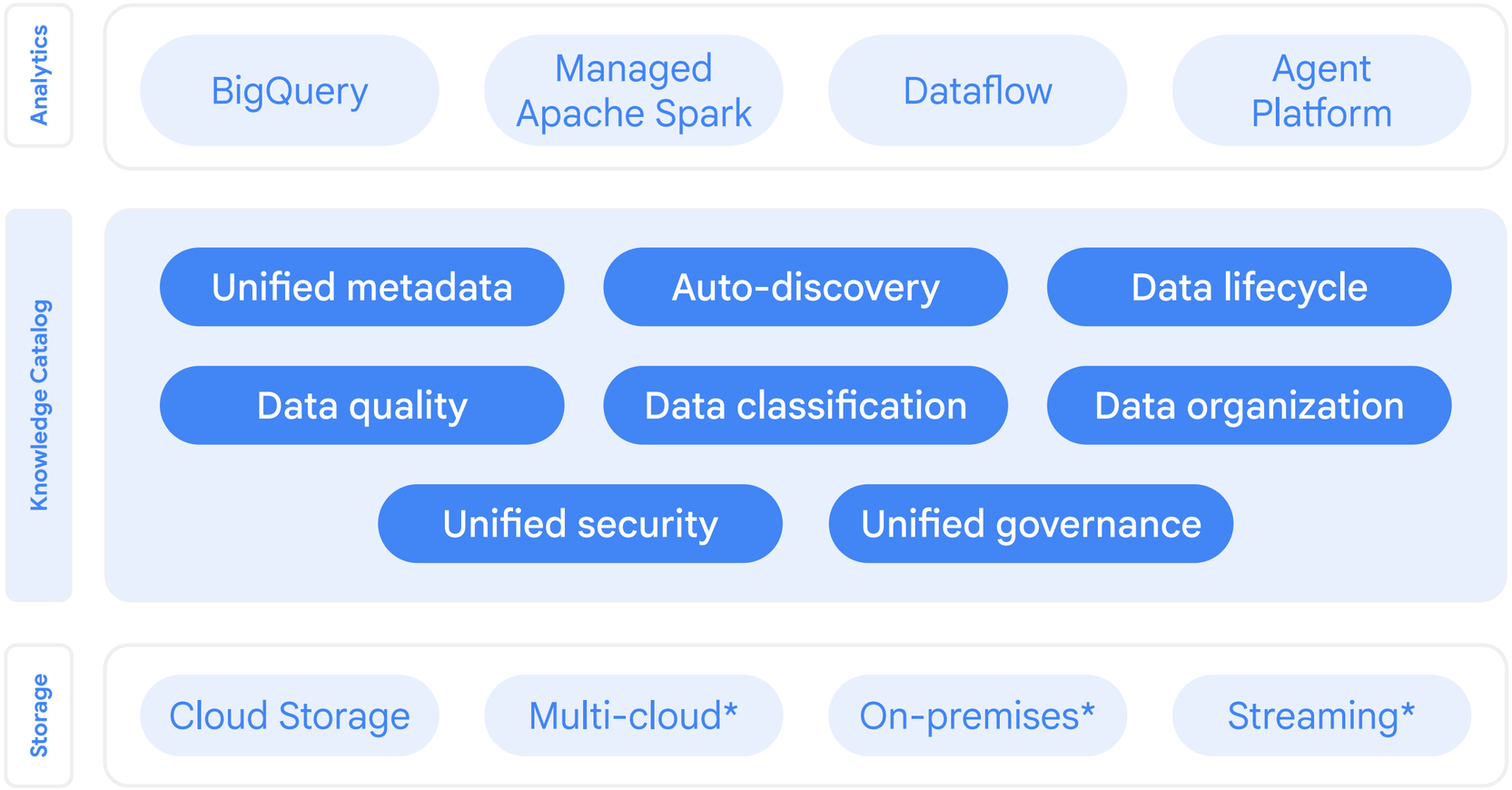
Data engineering tasks and components

01	The role of a data engineer
02	Data sources versus data syncs
03	Data formats
04	Storage solution options on Google Cloud
05	Metadata management options on Google Cloud
06	Share datasets using Analytics Hub



Proprietary + Confidential

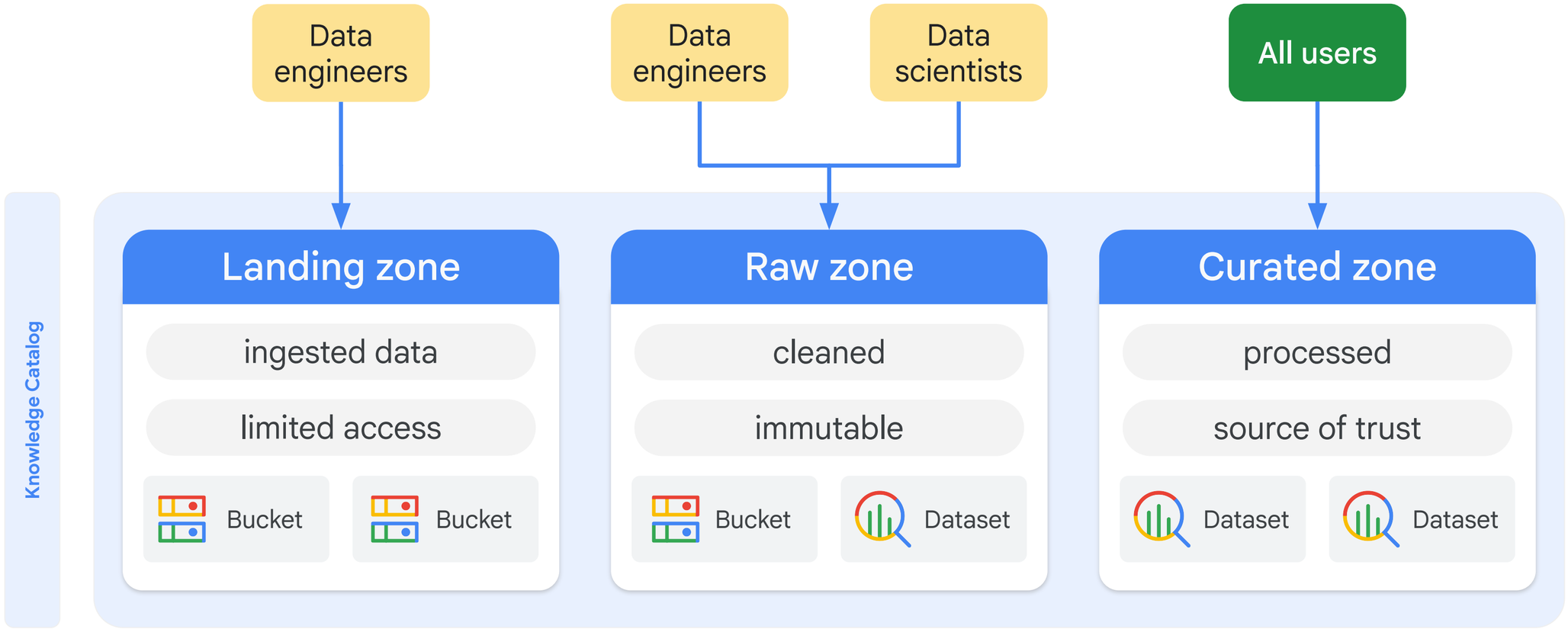
Centrally discover, manage, monitor, and govern distributed data with Knowledge Catalog



- Data to AI governance
- Data governance in BigQuery
- Insights and semantic search
- Data discovery with Data Catalog
- End-to-end data lineage

Proprietary + Confidential

Example: group and share your data based on readiness using Knowledge Catalog



Proprietary + Confidential

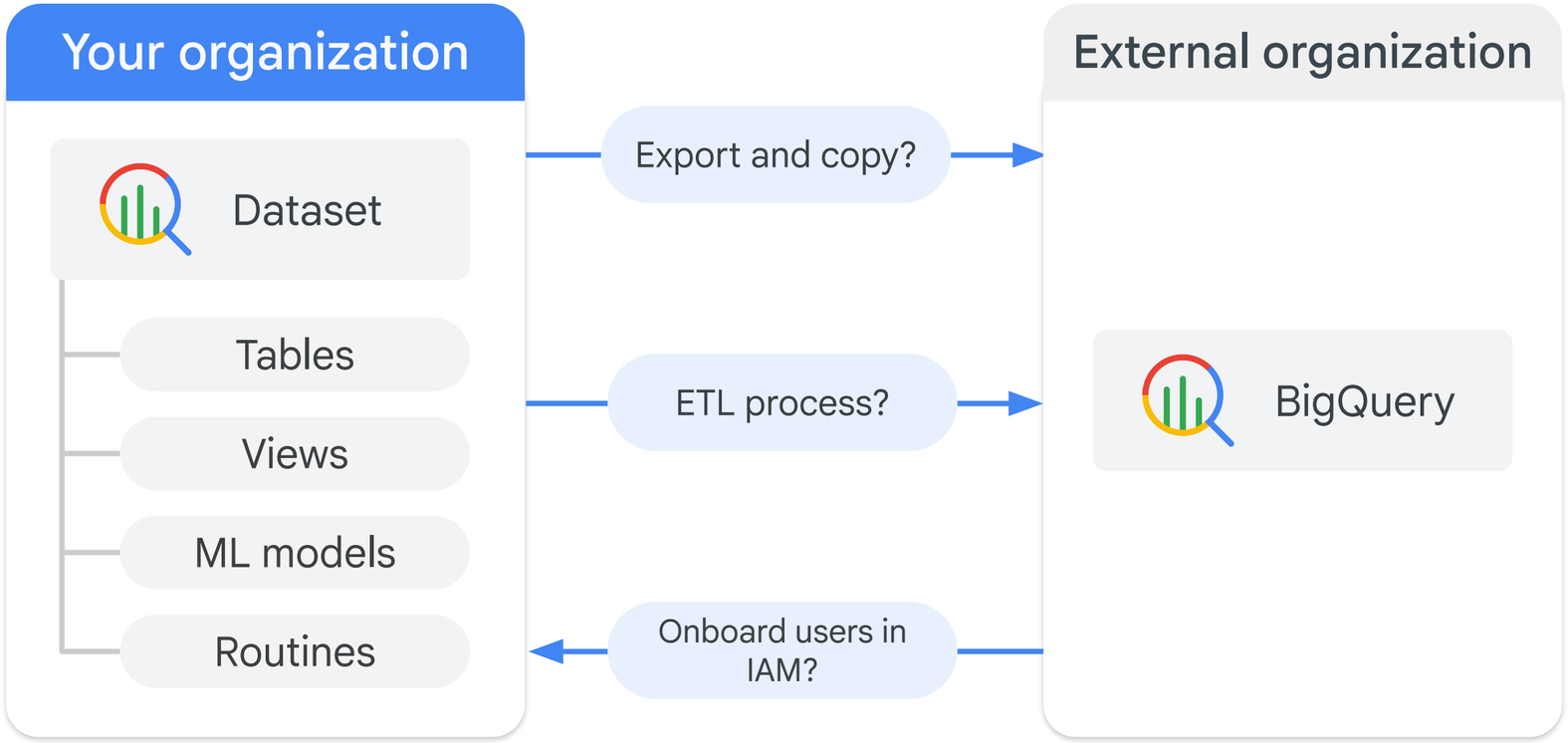
Data engineering tasks and components

01	The role of a data engineer
02	Data sources versus data syncs
03	Data formats
04	Storage solution options on Google Cloud
05	Metadata management options on Google Cloud
06	Share datasets using Analytics Hub



Proprietary + Confidential

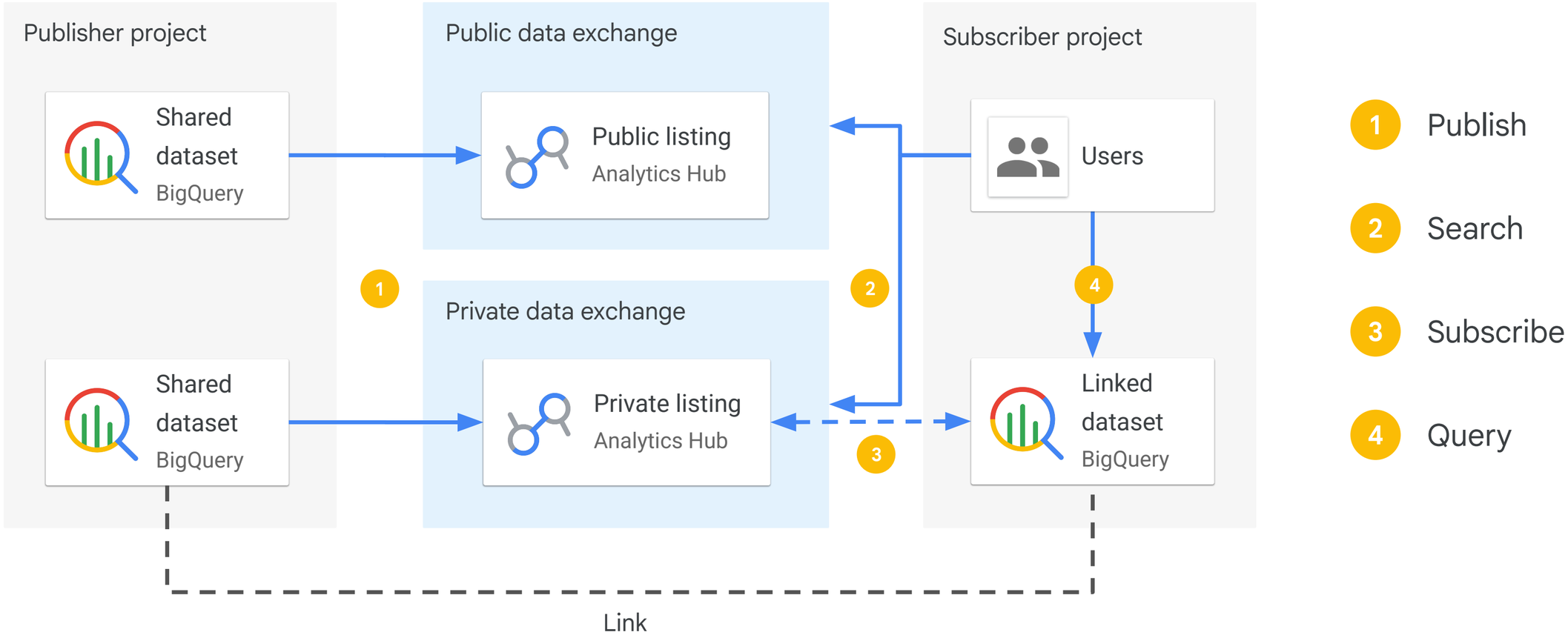
How about sharing data outside your organization? This is challenging



- Data freshness
- Pipeline management
- No data usage visibility
- Complex permissions

Proprietary + Confidential

Sharing data across organizations with Analytics Hub is easy



Google Cloud

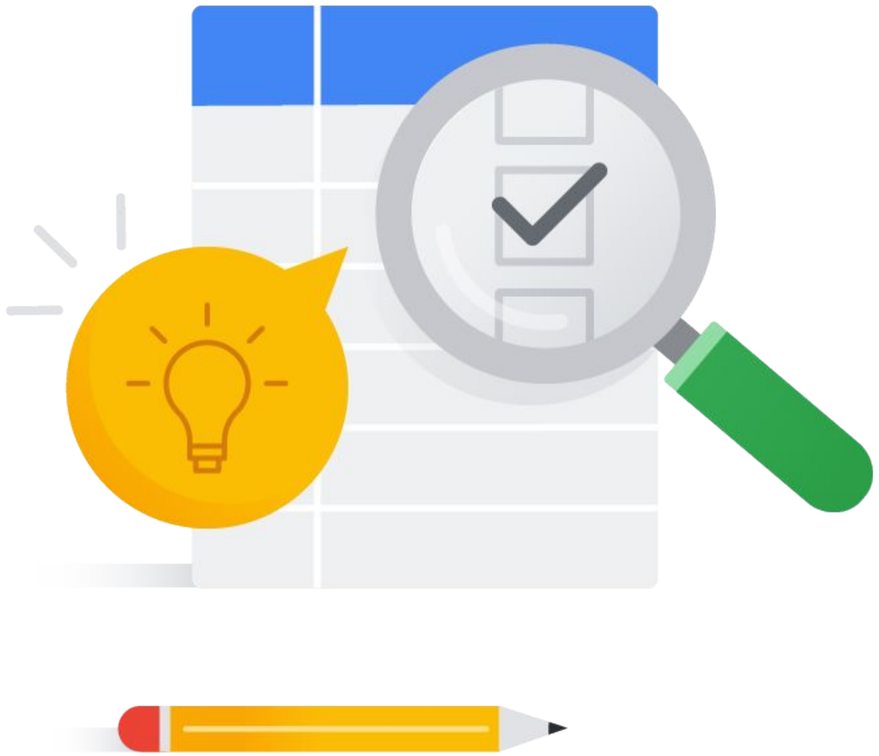
Proprietary + Confidential

Lab: Loading Data into BigQuery

30 min

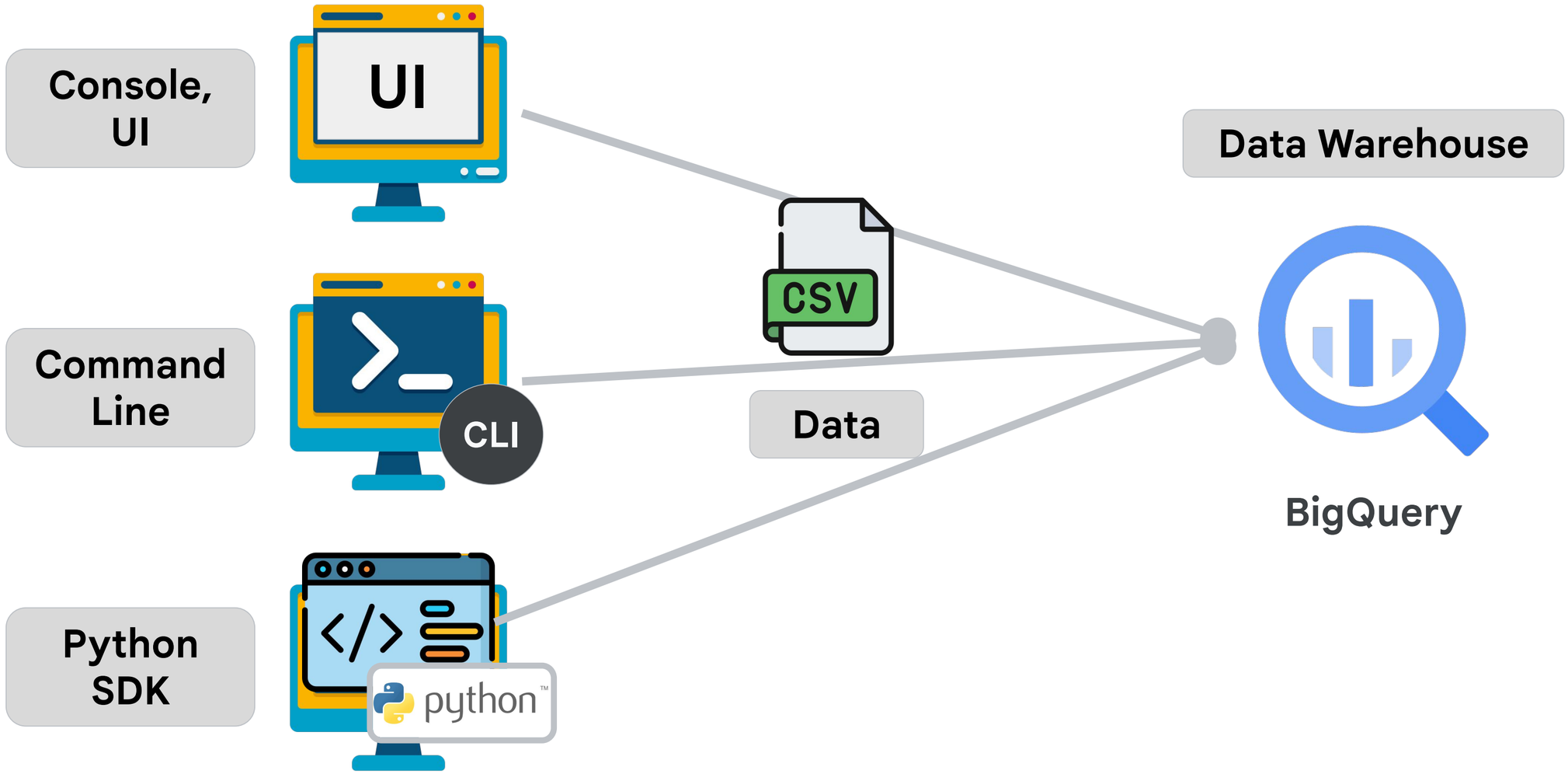
Learning objectives

- Load data into BigQuery from various sources.
- Load data into BigQuery using the CLI and the Google Cloud console.
- Use DDL to create tables.



Proprietary + Confidential

Lab: Loading Data into BigQuery




Google Cloud